BS-MS Curriculum at IISER Pune
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1. Historical Introduction to the Curriculum at IISER Pune

IISER Pune's BS-MS program started in 2006. For students admitted to the institute from 2006 to 2018, the curriculum of this five year dual degree program was notionally divided into three parts. The first part, comprising the first two years (or four semesters) had common and mandatory courses in all disciplines. The second part, comprising the third and fourth years (or four semesters), had advanced courses in every discipline, and the student was free to choose the courses that they were interested in. The only constraints were bounds on the number of credits per semester, and that the student was required to choose courses from at least two disciplines in the third year. The fifth year (or two semesters) was entirely devoted to a research project, and the student was expected to submit a thesis at the end of the fifth year and defend it to be eligible for the dual BS and MS degrees. The degrees of students graduating with this curriculum are not in a particular branch of science; rather, they are degrees in Science.

After about ten years of this curriculum being in place, sufficient data and feedback from students and faculty was available on it, following which an internal committee of faculty members called the Academic Advisory Committee was constituted in December 2017 to review the curriculum and suggest appropriate changes. After a survey was conducted amongst the faculty, senior students, and alumni to seek opinions and after several rounds of deliberations, the committee recommended that the curriculum be changed in the following way:

. In the first one and a half years (or three semesters), all students take common, compulsory courses that give them a broad background in the sciences. In the 3rd semester, besides the compulsory courses, students choose two elective courses.

a. In the 4th semester, all students take two compulsory courses and six elective courses. At this point, a student can begin to exercise any one of these three options: to major in one discipline; to major in one discipline and minor in another; to choose not to major or minor in any discipline. A student who wishes to major or minor has to take a certain number of prescribed courses in the 4th to 8th semesters (or five semesters) to fulfil the Major or Minor requirements.

b. In the fifth year (or two semesters), the student works on a research project and writes an MS thesis.

The changed structure of the BS-MS program was approved by the IISER Pune Senate during its 25th meeting in November 2018. Following that, the Curriculum Committee, an internal committee of faculty members representing all disciplines, met several times to deliberate on the nature of the courses in the new curriculum. Faculty meetings also took place in the various disciplines to discuss course structure and contents, and requirements for a Major and a Minor. An external review of the curriculum by distinguished academics and teachers was conducted on October 11, 2019 at IISER Pune. The present curriculum applies to BS-MS students admitted to IISER from the year 2019 onward. Minor modifications to the curriculum (changes in mandatory courses for ECS and HSS described below and credit requirements as detailed in the handbook of guidelines) were further introduced in 2021 and 2022.

This version of the handbook incorporates changes introduced till the batch 2024 (joining year 2024). All references to batches in the document are based on their year of joining.

2. Course Plan for a BS-MS Student

The table below outlines the number and nature of courses offered by different disciplines, and the number of courses that a typical BS-MS student would take during the five year program.

Discipline	**Sem 1**	**Sem 2**	**Sem 3**		**Sem 4**		**Sem 5 – 8**	**Sem 9 – 10**
	C	**C**	**C**	**E**	**C**	**E**	**E**	Research project and MS thesis
Biology	2	1	1	1	–	3	≥ 5	
Chemistry	1	2	1	1	–	3	≥ 5	
Mathematics	1	2	1	1	–	3	≥5	
Physics	2	1	1	1	–	3	≥5	
Earth and Climate Science (ECS)	1	1	1	1	–	3	≥5	
Humanities and Social Sciences (HSS)	–	1	1	–	–	–	≥3	
Foundation courses	2	–	–	–	2	–	–	
Total courses for a student	**9***	**8**	**6**	**2**	**2**	**6**	**6-7 per semester**	

C: Compulsory courses
E: Elective courses offered from which a student chooses an appropriate number as indicated in the last row
*For the 2019 and 2020 batches, there was no ECS course in Sem 1 and the total courses for a student in Sem 1 was 8.

With the introduction of the National Education Policy, exit options have been included to the program. In addition, summer internships have been allowed as credited courses (graded as Pass/Fail). These have been implemented starting from batch 2022. NPTEL-SWAYAM courses recommended by the respective disciplines can also be listed in the transcript as a Pass/Fail graded course, starting from the August 2024 semester.

Starting from batch 2022, to graduate with a BS-MS degree, the student has to obtain at least 220 credits, which are distributed in the following way: 99 (96 for 2019 and 2020 batches) in years one and two, ≥ 85 in years three and four (including summer project credits if opted for), and 36 in year 5 for the MS thesis. More information about credits for individual courses is provided below.

The following table summarises the credit completion requirements for the various batches. A course load of approximately 21 credits per semester is recommended for Years 3 and 4, to complete the credit requirements with a uniform distribution across the 4 semesters.

BATCH	Year 1&2	Year 3&4	Year 5	Total Credits	Major/ Minor	Remarks
2024	96	85+3	36	220	Yes	Sem 1, English course offered as remedial and not as a credited course
2023	99	85	36	220	Yes	
2022	99	85	36	220	Yes	2 + 2 summer credits introduced
2021	96+3*	84	36	219*	Yes	*ECS course introduced in Sem 1
2020	96	84	36	216	Yes	
2019	93/96	84/87	36	216	Yes	Lab course adjusted due to pandemic

2.1. Compulsory Courses

The table below outlines the compulsory courses ('C' for core or foundation courses) in the first three semesters and the two compulsory courses in the 4th semester.

Each course is worth 3 credits (3 contact hours, including two lecture hours and 1 tutorial session per week for theory courses and three contact hours for lab courses). Year 2 course instructors have the flexibility to utilise tutorial hours as lecture hours, if necessary.

The course design is such that a time commitment of 2.5 hours per credit is expected from the student, including contact hours and self-study hours.

The syllabi of all compulsory courses can be found here.

Semester 1					
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**
Biology	BI1113	Introductory Biology - I	3	1	Compulsory
Biology	BI1123	Experimental Biology	3	1	Compulsory
Chemistry	CH1113	Principles of Organic Chemistry	3	1	Compulsory
Earth and Climate Science	EC1213	Evolution of Earth and life	3	1	Compulsory
Mathematics	MT1113	Calculus - I	3	1	Compulsory
Physics	PH1113	Introductory Mechanics	3	1	Compulsory
Physics	PH1123	Physics Lab-I	3	1	Compulsory
Foundation courses	TD1113	Introduction to computing	3	1	Compulsory
Foundation courses	TD1123	Academic Communication Skills	3	1	Compulsory
Semester 2					
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**
Biology	BI1213	Introduction to Biomolecules	3	2	Compulsory
Chemistry	CH1213	Principles of Physical Chemistry	3	2	Compulsory
Chemistry	CH1223	General Chemistry Practicals I	3	2	Compulsory
Earth and Climate Science	EC1223	The Solid Earth	3	2	Compulsory
Humanities and Social Sciences	HS1223	Science and Society	3	2	Compulsory
Mathematics	MT1213	Calculus II	3	2	Compulsory
Mathematics	MT1223	Linear Algebra	3	2	Compulsory
Physics	PH1213	Introductory Electricity and Magnetism	3	2	Compulsory
Semester 3					
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**
Biology	BI2113	Ecology and Evolution	3	3	Compulsory
Chemistry	CH2113	Principles of Inorganic Chemistry	3	3	Compulsory
Earth and Climate Science	EC2113	Introduction to Climate Science	3	3	Compulsory

Humanities and Social Sciences	HS2123	Introduction to HSS	3	3	Compulsory
Mathematics	MT2113	Introduction to Probability	3	3	Compulsory
Physics	PH2113	Introductory Quantum Physics	3	3	Compulsory
Semester 4					
Foundation courses	TD2213	Thermodynamics	3	4	Compulsory
Foundation courses	TD2223	Data Analysis	3	4	Compulsory

*For the 2019 and 2020 batches, there was no ECS course in Sem 1 and the total number of compulsory courses for a student in Sem 1 was 8.

2.2. Elective Courses in the 3rd Semester

The table below lists the five elective courses in the 3rd semester of which a student is required to choose any two. Each course is worth 3 credits (3 contact hours per week).

The syllabi of the elective courses in Semester 3 can be found here.

Discipline	Code	Course	Credits	Semester	Compulsory/Elective
Biology	BI2123	Introduction to Biological Systems	3	3	Elective (E1)
Chemistry	CH2243	General Chemistry Practical II	3	3	Elective (E1)
Earth and Climate Science	EC2123	Landscapes and Their Evolution	3	3	Elective (E1)
Mathematics	MT2123	Advanced Linear Algebra*	3	3	Elective (E1)
Physics	PH2123	Mathematical Methods for Physics	3	3	Elective (E1)

None of the 3rd semester elective courses are a prerequisite for the courses in the 4th semester and are not a requirement for a Major or a Minor. Thus, the student has the flexibility to opt for courses in the 3rd semester without having to commit to a potential Major or Minor.

*Starting from batch 2024, the Semester 3 elective for Mathematics has been changed to 'Introduction to Proofs' (change introduced starting from the August 2025 semester). Advanced Linear Algebra was offered as the 3rd Semester elective for the batches 2019 – 2023.

2.3. Major and Minor Requirements for Each Discipline

From the 4th to the 8th semesters, students may opt for a Major and a Minor in disciplines of their choice. In any semester, no discipline requires that a student opt for more than five mandatory courses for being eligible for a Major. A student can, however, choose to take more than five courses in a discipline each semester. It should be noted that a student cannot opt for a Minor without also opting for a Major. Each discipline has different requirements regarding the number and nature of courses to be completed successfully by a student to qualify for a Major or a Minor in that discipline. The details are provided below.

It is to be noted that, like the courses of the 1st to the 3rd semesters, each course offered in the 4th semester is also worth 3 credits (3 contact hours per week). However, in the 5th to the 8th semesters, courses can be worth either 3 credits (2 contact hours per week) or 4 credits (3 contact hours per week). The reason for reduced contact hours in relation to credits is that a student is expected to do more homework and independent self-study in the higher semesters as the course content becomes more complex.

In addition to these courses, in the 5th to the 8th semesters, each discipline also offers semester-long "Lab Training/Theory Project" courses worth 3 credits (flexible number of contact hours) wherein a student can work in an experimental lab or take up a theory project with a faculty member of their choice. A student can register for at most one such project across disciplines per semester.

While the basic structure of the BS-MS program and the requirements for Major in all disciplines will remain unchanged, there may be small changes in the course contents, the semester when a course is offered, and new courses may be added.

The requirements of a Major / Minor in each discipline are summarised below. See the details in the sections below.

Summary of Discipline-wise Major/Minor requirements					
	Major	**Minor**	**Other remarks**	**Credit count: Major**	**Credit count: Minor**
Biology	Any 15 courses (BIXXXX), out of which 8 are 4-credit courses	Any 8 BIXXXX courses from Sem 4 - 8	Semester projects count towards a Major and/or Minor.	53 - 60 credits	24 - 32 credits
Chemistry	12 mandatory courses, and any 6 electives (CHXXXX) from Sem 4 - 8	Any 6 CHXXXX courses from Sem 4 - 8	Semester projects do not count towards Major and/or Minor; Only one advanced lab out of three counts for a Minor.	60 - 66 credits	18 - 24 credits
Data Science	Not offered currently	Not offered currently		Not applicable	Not applicable

Earth Sciences	3 mandatory courses, and any 12 electives (ECXXXX) from Sem 4 - 8	Any 6 ECXXXX theory courses from Sem 4 - 8	Maximum of one semester project counts towards Major; No lab or semester project courses are counted for a Minor.	50 - 59 credits	18 - 24 credits
Humanities and Social Sciences	Not offered currently	Any 6 HSXXXX courses from Sem 5 - 8	Only one semester project counts towards Minor.	Not applicable	18 - 24 credits
Mathematics	9 mandatory courses, and any 6 electives (MTXXXX) from Sem 4 - 8	Any 8 MTXXXX courses from Sem 4 - 8	Semester projects count towards a Major and/or Minor.	54 - 61 credits	24 - 32 credits
Physics	14 mandatory courses, and any 4 electives (PHXXXX) from Sem 4 - 8	Any 8 PHXXXX courses from Sem 4 - 8	Semester projects do not count towards Major and/or Minor.	65 - 69 credits	24 - 32 credits
Science Education	Not offered currently	Not offered currently		Not applicable	Not applicable

2.3.1 Biology

Major: At least 15 courses in Biology, of which a minimum of 8 need to be 4 credit courses

Minor: At least 8 courses in Biology from the list below.

Note: Semester Project courses **count** towards fulfilling the requirements for a Major and Minor. Any new course introduced in Semesters 5 – 8 with a 'BIXXXX' course code in future will also count towards Major/Minor, depending on the credits.

The syllabi of biology courses can be found here.

Semester 4 (Electives, E1)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**	**Major/Minor Requirement**	**Is this course Prerequisite for another course?**
Biology	BI2213	Cell Biology	3	4	Elective (E1)	Counted for Major/Minor	Yes
Biology	BI2223	Physiology	3	4	Elective (E1)	Counted for Major/Minor	
Biology	BI2233	Genetics	3	4	Elective (E1)	Counted for Major/Minor	
Semester 5 & 7 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major/Minor Requirement**	**Pre-Requisites**	**Is this course Prerequisite for another course?**
Biology	BI3124	Advanced Molecular Biology	4	5	Counted for Major/Minor		
Biology	BI3134	Bioinformatics	4	5,7	Counted for Major/Minor		
Biology	BI3144	Cellular Biophysics - I	4	5,7	Counted for Major/Minor		Yes
Biology	BI3154	Neurobiology - I	4	5,7	Counted for Major/Minor		Yes
Biology	BI3164	Plant Biology - I	4	5,7	Counted for Major/Minor		Yes
Biology	BI3174	Advanced Biochemistry - I	4	5,7	Counted for Major/Minor		
Biology	BI3184	Ecology - I	4	5,7	Counted for Major/Minor		
Biology	BI3194	Developmental Biology	4	5,7	Counted for Major/Minor		
Biology	BI3313	Semester Project	3	5	Counted for Major/Minor		
Biology	BI3323	Structural Biology	3	5,7	Counted for Major/Minor		
Biology	BI3344	Microbial Genetics	4	7	Counted for Major/Minor		
Biology	BI4113	Animal Physiology - II	3	7	Counted for Major/Minor	Animal Physiology I	
Biology	BI4123	Advanced Immunology	3	7	Counted for Major/Minor	Introductory Immunology	
Biology	BI4153	Undergraduate research experiments	3	5	Counted for Major/Minor		
Biology	BI4164	Computational Functional Genomics	4	7	Counted for Major/Minor		
Biology	BI4313	Semester Project	3	7	Counted for Major/Minor		
Semester 6 & 8 (Electives, E2)							

Discipline	Code	Course	Credits	Semester	Major/Minor Requirement	Pre-Requisites	Is this course Prerequisite for another course?
Biology	BI3214	Animal Physiology I	4	6,8	Counted for Major/Minor		Yes
Biology	BI3224	Introductory Immunology	4	6,8	Counted for Major/Minor	Cell Biology (E1 elective)	Yes
Biology	BI3234	Animal Behaviour	4	6,8	Counted for Major/Minor		
Biology	BI3254	Microbiology	4	6,8	Counted for Major/Minor		
Biology	BI3264	Mathematical & Computational Biology	4	6,8	Counted for Major/Minor		
Biology	BI3274	Chemical Ecology	4	6,8	Counted for Major/Minor		
Biology	BI3284	Advanced Biochemistry II	4	6,8	Counted for Major/Minor		
Biology	BI3333 (HS3173)	Disease and Discourse	3	6	Counted for Major/Minor		
Biology	BI3413	Physical Biochemistry	3	6,8	Counted for Major/Minor		
Biology	BI3424	Statistical learning and data science	4	6,8	Counted for Major/Minor		
Biology	BI3444	Genome Biology and Epigenetics	4	8	Counted for Major/Minor		
Biology	BI3613	Semester Project	3	6	Counted for Major/Minor		
Biology	BI4213	Applied Plant Biology	3	6,8	Counted for Major/Minor	Plant Biology	
Biology	BI4223	Neurobiology II	3	6,8	Counted for Major/Minor	Neurobiology I	
Biology	BI4233	Cellular Biophysics II	3	6,8	Counted for Major/Minor	Cellular Biophysics I	
Biology	BI4613	Semester Project	3	8	Counted for Major/Minor		

2.3.2 Chemistry

Major: At least 18 courses in Chemistry, of which 12 (see table below) are mandatory. The remaining 6 courses are as per the student's preference (Course code CHXXXX).

Minor: At least 6 courses in Chemistry (CHXXXX). Of these, **only one** out of the three advanced labs will count.

The syllabi of chemistry courses can be found here.

Semester 4 (Electives, E1)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**	**Major Minor Requirement**	**Is this course Prerequisite for another course?**
Chemistry	CH2213	Analytical Chemistry	3	4	Elective (E1)	**Mandatory for Major**; Counted for Minor	
Chemistry	CH2223	Principles of Organic Chemistry II	3	4	Elective (E1)	**Mandatory for Major**; Counted for Minor	
Chemistry	CH2233	Fundamentals of Molecular Spectroscopy	3	4	Elective (E1)	**Mandatory for Major**; Counted for Minor	Yes
Semester 5 & 7 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre-Requisites**	**Is this course Prerequisite for another course?**
Chemistry	CH3114	Physical Organic Chemistry	4	5,7	**Mandatory for Major**; Counted for Minor		
Chemistry	CH3124	Main Group Chemistry	4	5,7	**Mandatory for Major**; Counted for Minor		
Chemistry	CH3134	Symmetry and Group Theory	4	5,7	Counted for Major/Minor		
Chemistry	CH3143	Self-Assembly in Chemistry	3	5,7	Counted for Major/Minor		
Chemistry	CH3154	Chemical Equilibrium and Kinetics	4	5,7	**Mandatory for Major**; Counted for Minor		
Chemistry	CH3163*	Advanced Organic Chemistry Laboratory*	3	5,7	**Mandatory for Major**; Any one lab* counted for Minor.		
Chemistry	CH3313	Semester Project	3	5	**Not counted for Major or Minor**		

Chemistry	CH4114	Organic Synthesis - II	4	7	Counted for Major/Minor	Organic Synthesis I	
Chemistry	CH4124	Bioinorganic Chemistry	4	7	Counted for Major/Minor		
Chemistry	CH4134	Polymer Chemistry	4	7	Counted for Major/Minor		
Chemistry	CH4144	Statistical Thermodynamics	4	7	Counted for Major/Minor		
Chemistry	CH4153*	Advanced Physical Chemistry Laboratory*	3	7	**Mandatory for Major**; Any one lab* counted for Minor		
Chemistry	CH4164	Bioorganic Chemistry and Chemical Biology	4	7	Counted for Major/Minor		
Chemistry	CH4173	Solid State Chemistry	3	7	Counted for Major/Minor		
Chemistry	CH4184	Electrochemistry	4	7	Counted for Major/Minor		
Chemistry	CH4194	Fundamentals of Solution-State NMR Spectroscopy: Principles and Applications	4	7	Counted for Major/Minor		
Chemistry	CH4313	Semester Project	3	7	**Not counted for Major or Minor**		
Semester 6 & 8 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre-Requisites**	**Is this course Prerequisite for another course?**
Chemistry	CH3214	Quantum Chemistry	4	6,8	**Mandatory for Major**; Counted for Minor		
Chemistry	CH3224	Organic Synthesis-I	4	6,8	**Mandatory for Major**; Counted for Minor		Yes
Chemistry	CH3234	Transition metal Chemistry	4	6,8	**Mandatory for Major**; Counted for Minor		
Chemistry	CH3253*	Advanced Inorganic Chemistry Laboratory*	3	6,8	**Mandatory for Major**; Any one lab* counted for Minor		
Chemistry	CH3613	Semester Project	3	6	**Not counted for Major or Minor**		
Chemistry	CH4214	Organotransition Metal Catalysis	4	8	Counted for Major/Minor		
Chemistry	CH4224	Advanced Material Science	4	6,8	Counted for Major/Minor		
Chemistry	CH4233	Thermal Pericyclic and Photochemical Reactions	3	8	Counted for Major/Minor		
Chemistry	CH4243	Organometallic Chemistry	3	8	Counted for Major/Minor		
Chemistry	CH4254	Structural Methods and Analysis	4	8	Counted for Major/Minor		
Chemistry	CH4264	Advanced Molecular Spectroscopy	4	6,8	Counted for Major/Minor	Fundamentals of Molecular Spectroscopy	
Chemistry	CH4274	Medicinal Chemistry	4	8	Counted for Major/Minor		
Chemistry	CH4284	Chemistry for Alternative Energy	4	8	Counted for Major/Minor		
Chemistry	CH4323	Environmental Chemistry and Sustainability	3	8	Counted for Major/Minor		
Chemistry	CH4613	Semester Project	3	8	**Not counted for Major or Minor**		

Data Science

The Data Science discipline is currently not offering a major or minor.

Major: Not Applicable

Minor: Not Applicable

Students can take the following courses in data science, and do their MS thesis in this discipline. These courses are stated in the table below. Availability of courses may vary.

The syllabi of data science courses can be found here.

Semester 5 & 7 (Electives, E2)						
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Pre-Requisites**	
Data Science	DS3184	Data Assimilation	4	5,7		
Data Science	DS3313	Semester Project	3	5		
Data Science	DS4124	Computational Functional Genomics	4	7		

Data Science	DS4133	Deep Neural Networks	3	7		
Data Science	DS4313	Semester Project	3	7		
Semester 6 & 8 (Electives, E2)						
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Pre-Requisites**	
Data Science	DS3223	Causal inference	3	6,8		
Data Science	DS3244	Numerical Linear Algebra	4	6,8		
Data Science	DS3254	Systems and Implementation of Algorithms	4	6,8		
Data Science	DS3263	Signal Processing	3	6,8		
Data Science	DS3273	Image and video processing with deep learning	3	6,8		
Data Science	DS3613	Semester Project	3	6		
Data Science	DS4213	Natural Language Processing	3	6,8	Deep neural networks	
Data Science	DS4223	Generalized Linear Models and their Applications	3	6,8		
Data Science	DS4233	Time Series Analysis and Forecasting	3	6,8		
Data Science	DS4244	Bayesian theory and practice	4	6,8		
Data Science	DS4613	Semester Project	3	8		

In addition, an indicative list of courses from other departments that are relevant to Data Science is mentioned below.
This list and the availability of DS coded courses is subject to frequent changes.

BI3134 Bioinformatics
EC3183 Parameter Estimation and Inverse Theory
MT3194 Statistical Inference
MT3324 Markov Chain and its Application
MT4254 Stochastic Processes
MT3284 Topics: Mathematical Finance
MT3264 Algorithms
BI3424 Statistical learning and data science

Earth and Climate Sciences

For batch 2021 and onwards:

Major: A total of 15 courses. These should include 3 mandatory courses + 12 Electives (up to one semester project/lab project can be counted)

Minor: At least 6 theory courses in ECS.

The syllabi of Earth and Climate Science courses can be found here.

Semester 4 (Elective, E1)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**	**Major Minor Requirement**	**Is this course Prerequisite for another course?**
Earth and Climate Science	EC2213	Principles of Planetary Climate	3	4	Elective	**Mandatory for Major**; Counted for Minor	
Earth and Climate Science	EC2243	Atmosphere and Ocean Chemistry	3	4	Elective	Counted for Major/Minor	
Earth and Climate Science	EC2253	Introductory Seismology	3	4	Elective	Counted for Major/Minor	
Semester 5 & 7 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre Requisites**	**Is this course Prerequisite for another course?**
Earth and Climate Science	EC3124	Physics of the Atmosphere	4	5,7	Counted for Major/Minor		Yes
Earth and Climate Science	EC3154	Sedimentology and Stratigraphy	4	5	Counted for Major/Minor	Earth and Planetary Materials	

Earth and Climate Science	EC3164	Earth and Planetary Materials	4	5,7	**Mandatory for Major**; Counted for Minor		Yes
Earth and Climate Science	EC3174	Structural Geology and Tectonics	4	7	Counted for Major/Minor		Yes
Earth and Climate Science	EC3183	Parameter Estimation and Inverse Theory	3	5,7	Counted for Major/Minor		
Earth and Climate Science	EC3194	Paleobiology	4	5,7	Counted for Major/Minor		
Earth and Climate Science	EC3313*	Semester Project*	3	5	***Only one project counted for Major; Not counted for Minor**		
Earth and Climate Science	EC3323	Hydrology	3	5,7	Counted for Major/Minor		
Earth and Climate Science	EC3344 (DS3184)	Data Assimilation	4	5,7	Counted for Major/Minor		
Earth and Climate Science	EC4114	Atmosphere and Ocean Dynamics	4	7	Counted for Major/Minor	Geophysical Fluid Dynamics	
Earth and Climate Science	EC4123	Sedimentology & Palaeontology Lab	3	7	Counted for Major/Minor	Sedimentology and Stratigraphy,	
Earth and Climate Science	EC4134	Exploration Seismology	4	5,7	Counted for Major/Minor	Introductory Seismology	Yes
Earth and Climate Science	EC4144	Tropical Meteorology	4	7	Counted for Major/Minor	Geophysical Fluid Dynamics and Physics of Atmosphere	Yes
Earth and Climate Science	EC4153	Sequence Stratigraphy	3	7	Counted for Major/Minor	Sedimentology and Stratigraphy	Yes
Earth and Climate Science	EC4164	Igneous and Metamorphic Petrology	4	7	Counted for Major/Minor	Earth and Planetary Material, Geo and Cosmochemistry	Yes
Earth and Climate Science	EC4173	Igneous and Metamorphic Petrology Lab	3	7	Counted for Major/Minor	Igneous and Metamorphic Petrology	
Earth and Climate Science	EC4313*	Semester Project*	3	7	***Only one project counted for Major;Not counted for Minor**		
Semester 6 & 8 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre Requisites**	**Is this course Prerequisite for another course?**
Earth and Climate Science	EC3214	Geo and Cosmochemistry	4	6,8	Counted for Major/Minor	Earth and Planetary Materials	Yes
Earth and Climate Science	EC3224	Geophysical Fluid Dynamics	4	6,8	Counted for Major/Minor		Yes
Earth and Climate Science	EC3243	Analytical Geochemistry Lab	3	6,8	Counted for Major/Minor	Geo and Cosmochemistry	
Earth and Climate Science	EC3253	Introduction to Field Techniques	3	6,8	Counted for Major/Minor	Earth and Planetary Materials, Sedimentology and Stratigraphy, Structural Geology and Tectonics	
Earth and Climate Science	EC3264	Physical Oceanography	4	6,8	Counted for Major/Minor	Physics of the Atmosphere	Yes
Earth and Climate Science	EC3274	Geoelectromagnetic Exploration	4	6,8	Counted for Major/Minor		Yes
Earth and Climate Science	EC3284	Satellite Data Analysis & Image Processing	4	8	Counted for Major/Minor	Earth and Planetary Materials	
Earth and Climate Science	EC3293	Isotope Geochemistry	3	6,8	Counted for Major/Minor		
Earth and Climate Science	EC3414	Geodynamics-1	4	6,8	**Mandatory for Major**; Counted for Minor		
Earth and Climate Science	EC3613	Semester Project	3	6	**Only one project counted for Major; Not counted for Minor**		

Earth and Climate Science	EC4224	Climate Modelling	4	8	Counted for Major/Minor	Geophysical Fluid Dynamics, Either of the following: Physics of the Atmosphere / Physical Oceanography	
Earth and Climate Science	EC4243	Geological Field Training	3	8	Counted for Major	Sedimentology and Stratigraphy, Sequence Stratigraphy	
Earth and Climate Science	EC4254	Indian geology and resources	4	8	Counted for Major/Minor	Sedimentology and Stratigraphy, Geo and Cosmochemistry, Igneous and Metamorphic Petrology	
Earth and Climate Science	EC4274	Gravity and Magnetic Exploration	4	8	Counted for Major/Minor		Yes
Earth and Climate Science	EC4283	Geophysics Field Experiment and Data Analysis	3	8	Counted for Major	Exploration Seismology, Gravity and Magnetic Exploration, Geoelectromagnetic Exploration	
Earth and Climate Science	EC4613	Semester Project	3	8	**Only one project counted for Major; Not counted for Minor**		
Earth and Climate Science		Advanced Seismology		8	Counted for Major/Minor	Introductory seismology	

Humanities and Social Sciences

Major: Not applicable

Minor: Minimum 6 courses in HSS as per the student's preference, HSXXXX (not counting the compulsory courses in Semesters 2 and 3)

No more than 1 Lab Training/Theory Projects courses count towards fulfilling the requirements for a Minor.

The syllabi of Humanities and Social Science courses can be found here.

Semester 5 & 7 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre Requisites**	**Is this course Prerequisite for another course?**
Humanities and Social Sciences	HS3144	Select Key Political Concepts	4	5,7	Counted for Minor		
Humanities and Social Sciences	HS3153	Economics and Public Policy	3	5,7	Counted for Minor		
Humanities and Social Sciences	HS3173 (BI3333)	Disease and Discourse	3	5,7	Counted for Minor		
Humanities and Social Sciences	HS3183	History of India through 75 objects: Culture and Society	3	5,7	Counted for Minor		
Humanities and Social Sciences	HS3213	Introduction to Paninian Grammar	3	5,7	Counted for Minor		
Humanities and Social Sciences	HS3234	Science as Narrative in Literature and Cinema	4	5,7	Counted for Minor		
Humanities and Social Sciences	HS3253	Introduction to Political Ecology	4	5,7	Counted for Minor		
Humanities and Social Sciences	HS3313	Semester Project	3	5	**Counted for minor only once**		
Humanities and Social Sciences	HS3364	City and Cinema	4	7	Counted for Minor		
Humanities and Social Sciences	HS4313	Semester Project	3	7	**Counted for minor only once**		
Humanities and Social Sciences	TBA	Philosophy of the Constitution of India	3	5,7	Counted for Minor		
Humanities and Social Sciences	TBA	People, Culture, and Health	3	5,7	Counted for Minor		
Humanities and Social Sciences	TBA	Public Policy II			Counted for Minor		
Semester 6 & 8 (Electives, E2)							

Discipline	Code	Course	Credits	Semester	Major Minor Requirement	Pre Requisites	
Humanities and Social Sciences	HS3114	Disasters and Society	3	6,8	Counted for Minor		
Humanities and Social Sciences	HS3123	Evolution of Cinema	3	6	Counted for Minor		
Humanities and Social Sciences	HS3283	Development of Mathematical Astronomy in India	3	6,8	Counted for Minor		
Humanities and Social Sciences	HS3323	Philosophy of the Constitution of India	3	6,8	Counted for Minor		
Humanities and Social Sciences	HS3433	Introduction to Indian Writing in English	3	6,8	Counted for Minor		
Humanities and Social Sciences	HS3453	Public Policy I	3	6,8	Counted for Minor		
Humanities and Social Sciences	HS3463	Introduction to Archaeological Science	3	6,8	Counted for Minor		
Humanities and Social Sciences	HS3613	Semester Project	3	6	**Counted for minor only once**		
Humanities and Social Sciences	HS4613	Semester Project	3	8	**Counted for minor only once**		
Humanities and Social Sciences		Development Studies	3	6,8	Counted for Minor		

Mathematics

Major: Minimum 15 courses in Mathematics (courses with code of the form MTxxxx), of which 9 (see table below) are mandatory . The remaining 6 courses are as per the student's preference (courses with code of the form MTxxxx)

The courses offered in Semester 5 which are mandatory for a Major in Mathematics can be taken in Semester 7; similarly, the mandatory courses offered in Semester 6 can be taken in Semester 8.

Minor: Minimum 8 courses in Mathematics (courses with code of the form MTxxxx)

Note: Semester Projects courses **count** towards fulfilling the requirements for a Major and Minor.

The syllabi of Mathematics courses can be found here.

Semester 4 (Electives, E1)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**	**Major Minor Requirement**	**Is this course Prerequisite for another course?**
Mathematics	MT2213	Group Theory	3	4	Elective (E1)	**Mandatory for Major**; Counted for Minor	Yes
Mathematics	MT2223	Real Analysis I	3	4	Elective (E1)	**Mandatory for Major**; Counted for Minor	Yes
Mathematics	MT2233	Discrete Structures	3	4	Elective (E1)	**Mandatory for Major**; Counted for Minor	Yes
Semester 5 & 7 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre-Requisites**	
Mathematics	MT3114	Rings and Modules	4	5,7	**Mandatory for Major**; Counted for Minor	Group Theory	Yes
Mathematics	MT3124	Real Analysis II	4	5,7	**Mandatory for Major**; Counted for Minor	Real Analysis I	Yes
Mathematics	MT3134	Point Set Topology	4	5,7	**Mandatory for Major**; Counted for Minor	Real Analysis I	Yes
Mathematics	MT3144	Ordinary Differential Equations	4	5,7	**Mandatory for Major**; Counted for Minor	Real Analysis I	
Mathematics	MT3154	Graph Theory	4	5,7	**Mandatory for Major**; Counted for Minor	Discrete Structure	
Mathematics	MT3174	Fields and Galois theory	4	5,7	Counted for Major, Minor	Group Theory	
Mathematics	MT3184	Differential geometry and Lie groups	4	7	Counted for Major, Minor	Group Theory	
Mathematics	MT3313	Semester project	3	5	Counted for Major, Minor		
Mathematics	MT4114	Algebraic Topology	4	7	Counted for Major, Minor	Group Theory, Point Set Topology	
Mathematics	MT4124	Functional Analysis	4	7	Counted for Major, Minor	Measure Theory and Integration	Yes

Mathematics	MT4134	Probability	4	7	Counted for Major, Minor	Measure Theory and Integration	
Mathematics	MT4144	Representation theory of Finite Groups	4	7	Counted for Major, Minor	Advanced Linear Algebra, Algebraic Number Theory	
Mathematics	MT4313	Semester project	3	7	Counted for Major, Minor		
Mathematics	MT3194	Statistical Inference	4	5,7	Counted for Major, Minor		
Mathematics	MT3324	Markov Chain and its Application	4	5,7	Counted for Major, Minor		
Semester 6 & 8							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre-Requisites**	
Mathematics	MT3214	Complex Analysis	4	6,8	**Mandatory for Major**; Counted for Minor	Real Analysis I	
Mathematics	MT3224	Algebraic Number Theory	4	6,8	Counted for Major, Minor	Fields and Galois theory	
Mathematics	MT3234	Measure Theory and Integration	4	6,8	Counted for Major, Minor	Real Analysis II	Yes
Mathematics	MT3244	Calculus on Manifolds	4	6,8	Counted for Major, Minor	Point Set Topology, Real Analysis II	
Mathematics	MT3254	Coding Theory	4	6,8	Counted for Major, Minor	Discrete Structures (recommended, but not mandatory)	
Mathematics	MT3264	Algorithms	4	6,8	Counted for Major, Minor	Graph Theory	
Mathematics	MT3284	Topics: Mathematical Finance	4	6,8	Counted for Major, Minor		
Mathematics	MT3294	Topics: Computational number theory with Sage	4	6,8	Counted for Major, Minor		
Mathematics	MT3413	Topics: Symmetry and Lie Groups	3	6,8	Counted for Major, Minor		
Mathematics	MT3613	Semester Project	3	6	Counted for Major, Minor		
Mathematics	MT4214	Algebraic Geometry	4	8	Counted for Major, Minor	Rings and Modules	
Mathematics	MT4224	Fourier Analysis	4	8	Counted for Major, Minor	Functional Analysis	
Mathematics	MT4234	Riemannian Geometry	4	8	Counted for Major, Minor	Differential Geometry and Lie groups	
Mathematics	MT4244	Cryptography	4	8	Counted for Major, Minor	Group Theory	
Mathematics	MT4254	Stochastic Processes	4	8	Counted for Major, Minor	Probability	
Mathematics	MT4264	Partial Differential Equations	4	8	Counted for Major, Minor	Functional Analysis	
Mathematics	MT4294	Bayesian theory and practice	4	6,8	Counted for Major, Minor		
Mathematics	MT4613	Semester Project	3	8	Counted for Major, Minor		

Physics

Major: At least 18 courses in Physics, of which 14 are mandatory.

The remaining 4 courses are as per the student's preference (with course code PHXXXX).

Minor: At least 8 courses in Physics (with code PHXXXX)

Note: Semester Project courses **do not count** towards the requirements of major/ minor.

The syllabi of Physics courses can be found here.

Semester 4 (Electives, E1)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Compulsory/Elective**	**Major Minor Requirement**	**Is this course Prerequisite for another course?**
Physics	PH2213	Classical mechanics	3	4	Elective	**Mandatory for Major**; Counted for Minor	
Physics	PH2223	Thermal & Statistical Physics	3	4	Elective	**Mandatory for Major**; Counted for Minor	
Physics	PH2233	Physics Lab – II	3	4	Elective	**Mandatory for Major**; Counted for Minor	

Semester 5 & 7 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre Requisites**	**Is this course Prerequisite for another course?**
Physics	PH3114	Electrodynamics-I	4	5	**Mandatory for Major**; Counted for Minor		Yes
Physics	PH3124	Quantum Mechanics – I	4	5	**Mandatory for Major**; Counted for Minor		Yes
Physics	PH3134	Optics	4	5	**Mandatory for Major**; Counted for Minor		
Physics	PH3144	Electronics-I with lab	4	5	**Mandatory for Major**; Counted for Minor		
Physics	PH3153	Methods of Experimental Physics	3	5,7	Counted for Major, Minor		
Physics	PH3163	Mathematical Methods for Physics II	3	5,7	Counted for Major, Minor		Yes
Physics	PH3173	Physics Lab-III	3	5	Counted for Major, Minor		
Physics	PH3313	Semester Project	3	5	**Not counted for Major or Minor**		
Physics	PH4113	Condensed Matter Physics – II	3	7	Counted for Major, Minor	Condensed Matter Physics – I, Statistical Mechanics-I	
Physics	PH4123	Statistical Mechanics – II	3	7	Counted for Major, Minor	Statistical Mechanics – I	
Physics	PH4133	Quantum Field Theory-I	3	7	Counted for Major, Minor	Mathematical Methods for Physics II, Quantum Mechanics II, Group Theory in Physics	
Physics	PH4144	Physics Lab-V	4	7	**Mandatory for Major**; Counted for Minor		
Physics	PH4154	Nuclear and Particle Physics	4	7	**Mandatory for Major**; Counted for Minor	Quantum mechanics I	
Physics	PH4163	Astronomy and Astrophysics - I	3	7	Counted for Major, Minor		
Physics	PH4173	Fluid Dynamics	3	7	Counted for Major, Minor		
Physics	PH4183	Gravitation	3	5,7	Counted for Major, Minor		
Physics	PH4193	Physics at Nanoscales	3	7	Counted for Major, Minor	Quantum Mechanics I, Condensed Matter Physics I	
Physics	PH4273	Non-linear Dynamics	3	5,7	Counted for Major, Minor	Classical Mechanics	
Physics	PH4313	Semester Project	3	7	**Not counted for Major or Minor**		

Semester 6 & 8 (Electives, E2)							
Discipline	**Code**	**Course**	**Credits**	**Semester**	**Major Minor Requirement**	**Pre Requisites**	**Is this course Prerequisite for another course?**
Physics	PH3273	Electronics & Instrumentation	3	6,8	Counted for Major, Minor		
Physics	PH3214	Quantum Mechanics II	4	6	**Mandatory for Major**; Counted for Minor	Quantum Mechanics I	
Physics	PH3224	Condensed Matter Physics I	4	6	**Mandatory for Major**; Counted for Minor	Quantum Mechanics I, Thermal and Statistical Physics	Yes
Physics	PH3234	Statistical Mechanics I	4	6	**Mandatory for Major**; Counted for Minor	Thermal and Statistical Physics	Yes
Physics	PH3244	Physics Lab IV	4	6	**Mandatory for Major**; Counted for Minor		
Physics	PH3253	Group Theory in Physics	3	6,8	Counted for Major, Minor		Yes
Physics	PH3253	Group Theory in Physics	3	6,8	Counted for Major, Minor		
Physics	PH3264	Computational Physics	4	6,8	Counted for Major, Minor		
Physics	PH3264	Computational Physics	4	6,8	Counted for Major, Minor		
Physics	PH3273	Electronics & Instrumentation	3	6,8	Counted for Major, Minor	Electronics I	
Physics	PH3283	Electrodynamics II	3	6	Counted for Major, Minor	Electrodynamics I	
Physics	PH3293	Quantum Information and Quantum Computing	3	6,8	Counted for Major, Minor		
Physics	PH3613	Semester Project	3	6	**Not counted for Major or Minor**		
Physics	PH4213	Cosmology	3	8	Counted for Major, Minor		
Physics	PH4224	Atomic and Molecular Physics	4	8	**Mandatory for Major**; Counted for Minor	Quantum Mechanics I, Statistical Mechanics I	

Physics	PH4233	Physics Lab VI	3	8	Counted for Major, Minor		
Physics	PH4243	Advanced Gravitation	3	8	Counted for Major, Minor	Gravitation, Quantum Field Theory I	
Physics	PH4253	Quantum Optics	3	8	Counted for Major, Minor		
Physics	PH4253	Advanced Optics	3		Counted for Major, Minor	Optics, Quantum Mechanics I Electrodynamics I	
Physics Physics	PH4253 PH4263	Quantum Optics Astronomy and Astrophysics II	3 3	8 8	Counted for Major, Minor Counted for Major, Minor		
Physics	PH4283	Advanced Particle Physics	3	8	Counted for Major, Minor		
Physics	PH4293	Physics of Soft Matter	3	8	Counted for Major, Minor	Statistical Mechanics I	
Physics	PH4323	Quantum Information	3	7	Counted for Major, Minor	Quantum Mechanics I	
Physics	PH4413	Quantum Field Theory II	3	8	Counted for Major, Minor	Quantum Field Theory I	
Physics	PH4423	Experimental Techniques in High Energy Physics	3	8	Counted for Major, Minor		
Physics	PH4613	Semester Project	3	8	**Not counted for Major or Minor**		
Physics	PH5113	Advanced Classical Mechanics	3	7	Counted for Major, Minor	Classical Mechanics I	
Physics		Astrophysical Processes	3		Counted for Major, Minor		
Physics		Particle Physics – Standard Model	3		Counted for Major, Minor		
Physics	PH3273	Electronics & Instrumentation	3	6,8	Counted for Major, Minor		

A Note on prerequisites:

Instructors will assume that the student is familiar with the basic course (prerequisite course) while teaching the advanced course (the course which requires prerequisites). Physics majors will credit most of the prerequisites as a matter of course. Non-physics (Major) students may face course schedule/overload conflicts in crediting all the prerequisites. Non-physics (Major) students interested in physics courses for a Minor or otherwise may do so without the prerequisites after seeking permission from the instructor, and at their "own risk". The teaching of the advanced course will assume that all students are familiar with the basic courses listed as a prerequisite. Students should use this table to become familiar with the prerequisites before crediting the advanced course.

Science Education

The Science Education discipline is currently not offering a major or minor. The courses listed below offered for students in Semesters 7 and 8 of the BS-MS program will be counted as E2 electives.

Major: Not Applicable

Minor: Not Applicable

Students can take the following courses in science education, and do their MS thesis in this discipline. These courses are stated in the table below.

The syllabi of Science Education courses can be found here.

Discipline	Code	Course Name	Credits	Semester
Science Education	SE4113	The Cognitive Basis of Science	3	7
Science Education	SE4123	Science and the world	3	7
Science Education	SE4213	Pedagogy of Science	3	8
Science Education	SE4223	Role of Media and Models in Science Education	3	8
Science Education	SE4233	Philosophy of Education	3	8
Science Education	SE4313	Semester Project	3	7
Science Education	SE4613	Semester Project	3	8

2.4 Trans Disciplinary and NPTEL Courses

2.4.1 Trans Disciplinary Courses
Some courses that span multiple disciplines and focus on skill development for students are described below.

- TD 3112 and TD 4112: Summer Projects

A maximum of 2 credits is awarded for the internships during the summer period between the 2nd and 3rd years or 3rd and 4th years or both. The internship duration is recommended as 6-8 weeks. Internships must be in broad areas covered by the eight Departments at IISER Pune.

- TD 3113: Science and Technology Entrepreneurship

This course aims to provide students with foundational knowledge to pursue entrepreneurial ventures. It focuses on developing innovative ideas that address either unmet or existing needs in society, business, or among stakeholders.
The syllabus of this course can be found here.

2.4.2 Elective Courses from NPTEL-SWAYAM

IISER Pune has introduced NPTEL-SWAYAM courses for BSMS students, allowing students to enroll for credits and mentioned in the transcripts. Students may enroll in a maximum of one NPTEL course per semester, across disciplines. These courses are fully elective (E2) and optional. They do not fulfill major or minor requirements for any discipline but will contribute to the student's total completed credits (CC). Every semester, a list of courses approved by different disciplines is announced by the academic office.

IISER Pune evaluates these courses on a Pass/Fail/Incomplete basis, based on the scores received from NPTEL-SWAYAM.

2.4 Fifth Year Research Project

The fifth-year project, carried out over the 9th and 10th semesters, is worth 36 credits. The handbook for MS thesis details - the rules, and structure of MS thesis, and can be located here.

MS Thesis in HSS

To qualify for an MS Thesis in HSS, a student has to complete at least 4 electives in HSS in their 3rd and 4th years, no more than one of which can be a semester project course. Ideally, the semester project should be done under the faculty member who will potentially be the MS Thesis Supervisor/Expert.

A student is expected to draft a research proposal and bibliography at least three months before the starting date. This proposal and bibliography should be created and finalized in consultation with the potential MS Thesis supervisor/ advisor.